

ST ALOYSIUS (DEEMED TO BE UNIVERSITY), MANGALURU
MID Term Examination – Semester I – September 2026
BCA AI Augmented Software Systems
Subject: Programming with Python — ANSWER KEY
Subject Code: 3D102

Question 1

a) Trace Output [2 marks]

4.5

13.0

18

<class 'float'>

Justification: $p = 9$ is an int and $q = 4.0$ is a float. $p / 2$ uses true division, which always returns a float, giving 4.5. $p + q$ adds an int and a float, so Python implicitly converts p to float, giving 13.0. $r = "9"$ is a string; $\text{int}(r)$ explicitly converts it to the integer 9, so $p + \text{int}(r) = 9 + 9 = 18$ (an int). $\text{type}(p / 2)$ confirms the division result is a float.

b) Sample Program — Employee Payslip [8 marks]

```
name = input("Enter employee name: ")
emp_id = int(input("Enter employee ID: "))
basic = float(input("Enter basic salary: "))
hra = float(input("Enter HRA: "))
da = float(input("Enter DA: "))

gross = basic + hra + da
tax = 0.10 * gross
net = gross - tax

print(f"Name: {name}")
print(f"Employee ID: {emp_id}")
print(f"Gross Salary: {gross:.2f}")
print(f"Tax Deducted: {tax:.2f}")
print(f"Net Salary: {net:.2f}")
```

Sample run (basic=35000, HRA=8000, DA=4000): Gross Salary: 47000.00, Tax Deducted: 4700.00, Net Salary: 42300.00.

Question 2

a) Trace Output [2 marks]

3

2

289

True

Justification: $x // y$ is floor (integer) division: $17 // 5 = 3$. $x \% y$ is the remainder: $17 \% 5 = 2$. $x ** 2$ is exponentiation: $17 ** 2 = 289$. $x > y$ evaluates to True and $y > 0$ evaluates to True, so the and expression evaluates to True.

If $x = 20$ and $y = 6$: $x // y = 3$, $x \% y = 2$, $x ** 2 = 400$, and $x > y$ and $y > 0$ is still True.

b) Sample Program — Cab Fare Calculator [8 marks]

```
distance = float(input("Enter distance travelled (km): "))
passengers = int(input("Enter number of passengers: "))
is_night = input("Is it a night ride? (yes/no): ").strip().lower()

base_fare = distance * 15
passenger_charge = (passengers - 1) * 20 if passengers > 1 else 0
fare_before_surcharge = base_fare + passenger_charge

surcharge = 0
if is_night == "yes":
    surcharge = 0.15 * fare_before_surcharge

total_fare = fare_before_surcharge + surcharge

print(f"Base Fare: {base_fare:.2f}")
print(f"Passenger Charge: {passenger_charge:.2f}")
print(f"Night Surcharge: {surcharge:.2f}")
print(f"Total Fare: {total_fare:.2f}")
```

Sample run (distance=12, passengers=3, night ride=yes): Base Fare: 180.00, Passenger Charge: 40.00, Night Surcharge: 33.00, Total Fare: 253.00.

Question 3

a) Predicted Output [2 marks]

Status: Hot

Justification: temp = 38 is not greater than 40, so the first condition fails. It is greater than 30, so the elif temp > 30 branch is taken and status is set to "Hot".

b) Flow Chart / Pseudocode / Program — Income Tax Calculator [8 marks]

Flow chart outline (key steps):

- Start
- Input: annual income
- Decision: is income < 0? → Yes: display error message → End
- No → Decision: income <= 2,50,000? → Yes: tax = 0
- No → Decision: income <= 5,00,000? → Yes: tax = (income - 2,50,000) × 5%
- No → Decision: income <= 10,00,000? → Yes: tax = 12,500 + (income - 5,00,000) × 20%
- No → tax = 12,500 + 1,00,000 + (income - 10,00,000) × 30%
- Compute cess = tax × 4%; total tax = tax + cess
- Display total tax payable
- End

Pseudocode:

```
START
READ income
IF income < 0 THEN
    PRINT "Error: income cannot be negative"
ELSE
```

```

IF income <= 250000 THEN tax = 0
ELSE IF income <= 500000 THEN tax = (income-250000)*0.05
ELSE IF income <= 1000000 THEN tax = 12500+(income-500000)*0.20
ELSE tax = 12500+1000000+(income-1000000)*0.30
cess = tax * 0.04
total_tax = tax + cess
PRINT total_tax

```

END

Python Program:

```

income = float(input("Enter annual income: "))

if income < 0:
    print("Error: Income cannot be negative.")
else:
    if income <= 250000:
        tax = 0
    elif income <= 500000:
        tax = (income - 250000) * 0.05
    elif income <= 1000000:
        tax = (250000 * 0.05) + (income - 500000) * 0.20
    else:
        tax = (250000 * 0.05) + (500000 * 0.20) + (income - 1000000) * 0.30

    cess = tax * 0.04
    total_tax = tax + cess

    print(f"Tax before cess: {tax:.2f}")
    print(f"Cess (4%): {cess:.2f}")
    print(f"Total Tax Payable: {total_tax:.2f}")

```

Sample run (income=800000): Tax before cess: 72500.00, Cess (4%): 2900.00, Total Tax Payable: 75400.00.

Question 4

a) Trace Output [2 marks]

Pass

Justification: marks = 55 (≥ 40) and attendance = 80 (≥ 75), so both conditions of the first if are True, giving "Pass".

If marks = 35 and attendance = 90: marks ≥ 40 is False, so the if fails. In the elif, marks ≥ 40 is False but attendance ≥ 75 is True, so the or condition is True — output is "Condonation".

b) Sample Program — Loan Eligibility Checker [8 marks]

```

age = int(input("Enter age: "))
income = float(input("Enter monthly income: "))
credit_score = int(input("Enter credit score: "))
experience = int(input("Enter years of employment: "))

criteria_met = 0
if 21 <= age <= 60:
    criteria_met += 1
if income >= 25000:
    criteria_met += 1
if credit_score >= 700:
    criteria_met += 1
if experience >= 2:
    criteria_met += 1

if criteria_met == 4:
    category = "Excellent"
elif criteria_met in (2, 3):
    category = "Good"
else:
    category = "Poor"

print(f"Criteria met: {criteria_met}")
print(f"Eligibility Category: {category}")

```

Sample run (age=30, income=28000, credit_score=720, experience=3): Criteria met: 4, Eligibility Category: Excellent.

Question 5

a) Trace Output [2 marks]

2 1 2

3 1 5

3 2 11

Final total = 11

Justification: i and j both range from 1 to 3. Whenever $i == j$, break immediately exits the inner for loop (not just that iteration), skipping all remaining j values for that i. For $i=1$: $j=1$ triggers $i==j$ immediately, so the inner loop breaks right away — nothing is printed for $i=1$. For $i=2$: $j=1$ gives $\text{total} += 2*1 = 2$, printed as (2, 1, 2); $j=2$ triggers break. For $i=3$: $j=1$ gives $\text{total} += 3*1 = 5$, printed as (3, 1, 5); $j=2$ gives $\text{total} += 3*2 = 11$, printed as (3, 2, 11); $j=3$ triggers break. Final total = 11.

b) Sample Program — Weekly Temperature Analysis [8 marks]

```

total_temp = 0
heatwave_days = 0
hottest_day = 1
hottest_temp = float('-inf')

for day in range(1, 8):
    c = float(input(f"Enter max temperature for day {day} (C): "))
    f = c * 9 / 5 + 32
    print(f"Day {day}: {c:.2f}C = {f:.2f}F")
    total_temp += c
    if c > 40:
        heatwave_days += 1
    if c > hottest_temp:
        hottest_temp = c
        hottest_day = day

average_temp = total_temp / 7
print(f"Average Temperature: {average_temp:.2f}C")
print(f"Heatwave Days (>40C): {heatwave_days}")
print(f"Hottest Day: Day {hottest_day} with {hottest_temp:.2f}C")

```

Sample run (daily max temps: 32, 35, 41, 38, 29, 42, 30): Average Temperature: 35.29°C, Heatwave Days (>40°C): 2, Hottest Day: Day 6 with 42.00°C.